

Module II: Inheritance

Sr	Question	BTL	CO
1.	Explain the concept of inheritance in C++ with a suitable example.	L2	F015-2
2.	Explain different types of inheritance in C++ with diagrams.	L2	F015-2
3.	Describe the relationship between access specifiers and visibility modes in inheritance.	L2	F015-2
4.	Explain single inheritance with an example.	L2	F015-2
5.	Explain multilevel inheritance with a suitable example.	L2	F015-2
6.	Explain multiple inheritance and its challenges.	L2	F015-2
7.	Explain hierarchical inheritance with an example.	L2	F015-2
8.	Explain hybrid inheritance and its challenges.	L2	F015-2
9.	Explain function overriding with a suitable example.	L2	F015-2
10.	Explain virtual base classes and their need in C++.	L2	F015-2
11.	Explain abstract classes and pure virtual functions in C++.	L2	F015-2
12.	Differentiate between multilevel and multiple inheritance.	L2	F015-2
13.	Analyze how visibility modes affect data accessibility in derived classes.	L4	F015-2
14.	Analyze the role of virtual base classes in resolving ambiguity in hybrid inheritance.	L4	F015-2

15.	Write a C++ program to demonstrate multilevel inheritance using a real-life example.	L3	F015-2
16.	Write a program to implement multiple inheritance and access members of both base classes.	L3	F015-2
17.	Develop a C++ program to demonstrate hierarchical inheritance.	L3	F015-2
18.	Write a program to demonstrate hybrid inheritance using virtual base class.	L3	F015-2
19.	Write a C++ program to pass parameters to the base class constructor and display values.	L3	F015-2
20.	Write a program to demonstrate function overriding using base and derived class.	L3	F015-2
21.	Develop a program to show the use of friend class in inheritance.	L3	F015-2
22.	Write a C++ program to demonstrate nested classes and access their members.	L3	F015-2
23.	Create a class Vehicle and derive Car and Bike classes to demonstrate inheritance and overriding.	L3	F015-2

24.	Write a program to demonstrate ambiguity problem in multiple inheritance and resolve it using scope resolution.	L4	F015-2
25.	Develop a program using abstract class Shape and derive classes Circle and Rectangle to calculate area.	L3	F015-2
26.	Write a program to demonstrate constructor execution order in multilevel inheritance.	L4	F015-2
27.	Analyze and implement a scenario where base class pointer accesses derived class object using overriding (runtime behavior explanation required).	L4	F015-2
28.	Design a class Employee with base class Salary and derived class Bonus. Pass values to base constructor and compute total salary.	L3	F015-2
29.	Explain accessibility of members in different inheritance modes	L2	F015-2

30.	Create base class "BankAccount" with function calculateInterest(). Override it in "SavingsAccount" and "CurrentAccount" classes.	L3	F015-2
31.	Design a smart home devices program where SmartDevice is the base class, containing common function like turning ON/OFF. SmartBulb and SmartSpeaker inherit from SmartDevice. A SmartHomeAssistant (like Alexa) inherits from both SmartSpeaker and SmartBulb to have both functionalities.	L3	F015-2
32.	Design a Banking System where: A class BankAccount stores account details like account number, balance, and PIN (private). Inside it, create a nested class Transaction that handles deposit and withdrawal operations. The Transaction class should inherit from a base class SecurityCheck that verifies basic transaction rules. Create a friend class Auditor that can access private details of BankAccount for auditing purposes.	L3	F015-2

Module III: Functions and Polymorphism

Sr	Question	BTL	CO
1.	Explain inline functions in C++ with suitable example.	L2	F015-3
2.	Explain friend function and its characteristics.	L2	F015-3
3.	Explain static data members and static member functions.	L2	F015-3
4.	Define polymorphism in C++ and explain its types.	L2	F015-3
5.	Explain function overloading with suitable example.	L2	F015-3

Module III: Functions and Polymorphism

6.	Explain constructor overloading in C++.	L2	F015-3
7.	Explain operator overloading in C++ with syntax.	L2	F015-3
8.	Differentiate between unary and binary operator overloading.	L2	F015-3
9.	Explain overloading of insertion (<<) and extraction (>>) operators.	L2	F015-3
10.	Explain virtual functions with suitable example.	L2	F015-3
11.	Explain pure virtual functions and abstract classes.	L2	F015-3
12.	Enlist rules of Virtual functions.	L2	F015-3
13.	When do we make a virtual function “Pure”? Explain the implications of making a function a pure virtual function.	L3	F015-3
14.	Differentiate between compile-time and run-time polymorphism.	L2	F015-3
15.	Explain the advantages of function overloading.	L2	F015-3
16.	Explain the need for virtual functions in C++.	L2	F015-3
17.	Analyze the difference between static and non-static members.	L4	F015-3

18.	Write a C++ program to demonstrate inline function for calculating square of a number.	L3	F015-3
19.	Write a program to demonstrate friend function accessing private members of a class.	L3	F015-3
20.	Write a C++ program to demonstrate use of static data member and static function.	L3	F015-3
21.	Write a program to implement function overloading for addition of two and three numbers.	L3	F015-3
22.	Develop a program to demonstrate constructor overloading.	L3	F015-3
23.	Write a program to overload unary operator (++) to increment object value.	L3	F015-3

24.	Write a program to overload binary operator (+) to add two objects.	L3	F015-3
25.	Write a C++ program to overload << and >> operators for a class Student.	L3	F015-3
26.	Write a program to demonstrate runtime polymorphism using virtual function.	L3	F015-3
27.	Develop a program using pure virtual function and abstract class Shape.	L3	F015-3

28.	Analyze and implement function overloading with different data types and explain ambiguity if any.	L4	F015-3
29.	Create a base class Vehicle with a function start(). Derive classes Car and Bike and override the start() function in each class to display different starting mechanisms. Write a C++ program to show how overridden functions behave for different objects.	L4	F015-3
30.	Write program to demonstrate the difference between compile-time and runtime polymorphism using suitable example for each.	L4	F015-3
31.	Develop a program where operator overloading is used to compare two objects (greater than).	L3	F015-3
32.	Analyze the working of virtual function using base class pointer and derived class object.	L4	F015-3
33.	Design a class Complex and overload + and << operators to perform addition and display of complex numbers.	L3	F015-3
34.	Friend function acts as bridge between two classes. Justify using example.	L3	F015-3
35.	Write a program to count number of objects using static member.	L3	F015-3
36.	How does the compiler differentiate overloaded functions? Explain using example.	L3	F015-3
37.	Compare operator overloading using: • member function • friend function	L3	F015-3

38.	Implement a class “Complex” and overload “+” operator for addition and “-” operator for subtraction.	L3	F015-3
39.	Design an abstract class “Vehicle” with pure virtual function “start()”. Implement Car and Bike classes from Vehicle Class.	L3	F015-3
40.	Create a base class Employee with a function calculateSalary(). Derive classes FullTimeEmployee and PartTimeEmployee and override the calculateSalary() function based on their salary calculation rules. Write a C++ program to demonstrate function overriding.	L3	F015-3
41.	Smart Home Control Panel Your team is designing software for a Smart Home Control System. One of the components is a Temperature Controller, which continuously monitors and reports current temperature readings. You need: A function that converts temperature from Celsius to Fahrenheit very frequently. A Power Management Unit that needs to access battery status stored privately in the TemperatureController class for making energy-saving decisions. A way to track how many total times the temperature was checked, regardless of how many objects were created.	L3	F015-3

Module IV: Advanced Concepts in C++

Sr	Question	BTL	CO
1.	Explain the concept of exception handling in C++.	L2	F015-4
2.	Explain the basic structure of try, catch, and throw with example.	L2	F015-4
3.	What are exceptions? List different types of exceptions in C++.	L2	F015-4
4.	Explain simple exception handling with suitable example.	L2	F015-4
5.	Explain multiple catch blocks with example.	L2	F015-4
6.	Explain the advantages of exception handling.	L2	F015-4
7.	Differentiate between syntax errors and runtime errors.	L2	F015-4
8.	Explain the use of catch(...) using example.	L2	F015-4
9.	Explain file handling in C++ and its importance.	L2	F015-4
10.	Explain types of files: sequential and random access files.	L2	F015-4
11.	Explain different file stream classes in C++.	L2	F015-4
12.	Explain file opening modes in C++.	L2	F015-4
13.	Explain file pointer operations seekg and seekp.	L2	F015-4
14.	Explain how to open and close files in C++.	L2	F015-4
15.	Explain how end-of-file (EOF) is detected in file handling.	L2	F015-4
16.	Explain following file modes using example: (i) ios::in (ii) ios::out (iii) ios::app (iv) ios::binary	L2	F015-4
17.	Analyze the use of multiple catch blocks in handling different exceptions.	L4	F015-4
18.	Analyze differences between sequential and random file access.	L4	F015-4
19.	Write a C++ program to store student records in a file, read and display them.	L3	F015-4

20.	Write a C++ program to demonstrate simple exception handling using try and catch.	L3	F015-4
21.	Write a program to handle division by zero using exception handling.	L3	F015-4
22.	Write a program demonstrating multiple catch blocks for different data types.	L3	F015-4
23.	Develop a program to throw and catch user-defined exceptions.	L3	F015-4
24.	Write a program to demonstrate nested try-catch blocks.	L3	F015-4
25.	Write a C++ program to create a file and write data into it.	L3	F015-4
26.	Write a program to read data from a file and display it on screen.	L3	F015-4
27.	Write a program to append data into an existing file.	L3	F015-4
28.	Develop a program to count the number of characters, words, and lines in a file.	L3	F015-4
29.	Write a program to copy contents from one file to another.	L3	F015-4

30.	Analyze and implement exception handling for invalid user input (e.g., negative values).	L4	F015-4
31.	Write a program to demonstrate random file access using seekg() and tellg().	L3	F015-4
32.	Develop a program to store and retrieve student records using file handling.	L3	F015-4
33.	Write a program using multiple catch blocks to handle 3 different exceptions.	L3	F015-4
34.	Analyze how exception handling improves program reliability with suitable examples.	L4	F015-4

35.	Design a program that combines file handling and exception handling to manage file errors (file not found, read/write error).	L3	F015-4
36.	Analyze and write what will happen in the following situations: i. An exception is thrown outside a try block. ii. No catch handler matches the type of exception thrown. iii. Several handlers match the type of exception thrown. iv. A catch handler throws an exception. v. A function throws an exception of type not specified in the specification list. vi. An exception rethrown within a catch block.	L3	F015-4
37.	Bank Withdrawal System Problem: Design a program that allows a user to withdraw money from their account. If the user tries to withdraw more than their current balance, throw an exception. Also, prevent negative withdrawal amounts. Exception ideas: Use invalid_argument or logic_error for both cases.	L3	F015-4
38.	Write a C++ program to read a file and count Character, Word, Line using class, object, file handling	L3	F015-4
39.	Write a C++ program to read a file and count the number of vowels, digits, and special characters using class, object, and exception handling.	L3	F015-4
40.	Write a C++ program to Create a file, write data into it, and then copy its contents into another file	L3	F015-4
41.	Write a C++ program to store student records in a file and display them	L3	F015-4
42.	Write a C++ program to append data into an existing file and display updated contents	L3	F015-4
43.	Write a C++ program to read a file and display each line along with its line number	L3	F015-4
44.	Write a C++ program to read a file and display its contents in reverse order	L3	F015-4